

Paper Summary

Diagnosing and Re-learning for Balanced Multimodal Learning

English Summary with Figures and Tables

Core Topic

Balanced multimodal learning, modality imbalance, modality capacity limitations, and soft encoder re-initialization.

Item	Details
Paper Type	Multimodal Learning / Modality Imbalance
Main Task	Balancing the learning process of different modalities
Main Modalities in Experiments	Audio, Vision, RGB, Optical Flow, and Text
Proposed Method	Diagnosing & Re-learning
Why It Matters for Our Project	It helps explain how an audio-face prototype can avoid over-relying on one modality when the other is noisy or weak.

1. Paper Overview

This paper addresses the imbalanced multimodal learning problem, where a multimodal model may prefer or over-rely on specific modalities while under-learning others. For example, in an audio-visual model, the model may rely more on audio and ignore visual cues, or vice versa.

The authors argue that previous balancing methods often make a risky assumption: the modality with worse prediction performance always needs more training. However, some modalities may be weak by nature because they contain little label-related information or too much noise. Forcing the model to learn from such modalities can make it memorize noise and reduce generalization.

Main Idea

Instead of blindly emphasizing weak modalities, the paper diagnoses the learning state of every modality and then softly re-initializes each uni-modal encoder according to its learning state.

2. Problem Statement

Existing multimodal imbalance methods usually focus on the performance gap between modalities. They may suppress dominant modalities or emphasize weak modalities. The paper highlights two key limitations:

- They ignore the intrinsic capacity limitation of some modalities, especially when a modality is noisy or scarcely informative.
- They often focus only on the worse-learned modality and may neglect or even harm the well-learned modality.
- They may fail in real-world scenarios where one modality is unreliable, noisy, or not strongly related to the target label.

3. Proposed Solution: Diagnosing & Re-learning

The proposed method has two main stages: Diagnosing and Re-learning.

3.1 Diagnosing

The method estimates the learning state of each modality by measuring the separability of its uni-modal representation space. The authors use clustering purity to compare the representation quality on the training set and validation set. A large purity gap indicates that the modality may be well-learned or even over-trained.

3.2 Re-learning

After diagnosing each modality, the method softly re-initializes the corresponding uni-modal encoder. This means that the encoder parameters are partially moved back toward their initial values, rather than being fully reset. The re-initialization strength depends on the diagnosed learning state.

This helps reduce excessive reliance on dominant modalities, prevent over-training, and avoid memorizing noise in scarcely informative modalities.

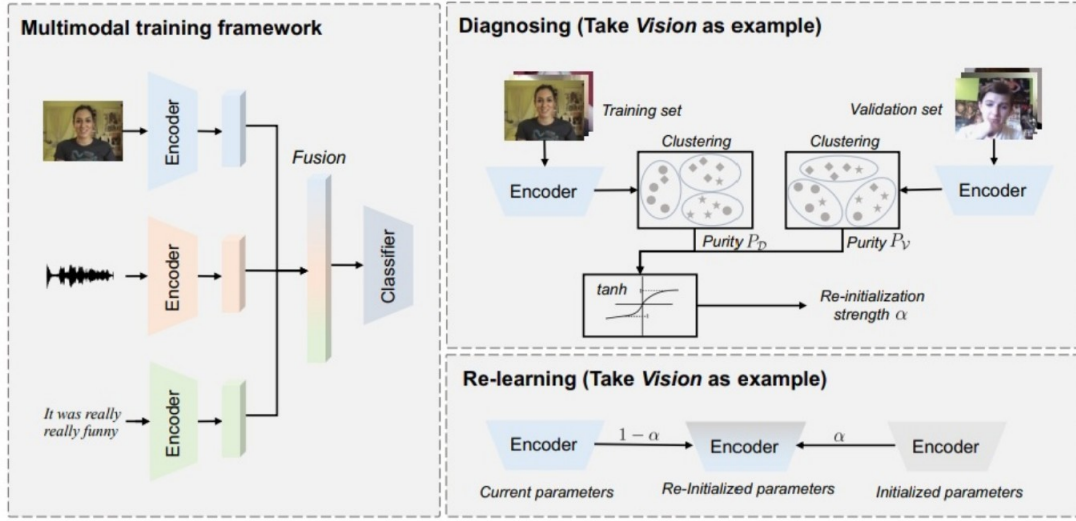


Fig. 2: Illustration of multimodal framework and the proposed Diagnosing & Re-learning method.

Figure 2. The general multimodal training framework and the proposed Diagnosing & Re-learning method.

4. Why This Method Is Important

The method is important because it balances all modalities at once. It does not simply force the weak modality to learn more. Instead, it considers the learning state and potential limitations of every modality. This makes the method useful for real-world multimodal systems where some modalities may be noisy, missing, or naturally less informative.

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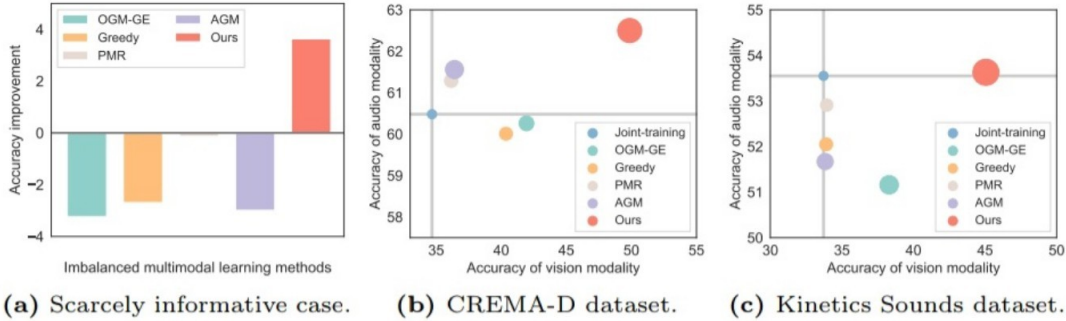


Fig. 1: (a): Scarcely informative modality case. It shows the accuracy improvement compared with the joint-training baseline. Only our method has a positive performance improvement. **(b)&(c): Uni-modal encoder quality evaluation and comparison.** The uni-modal evaluation (Acc audio and Acc vision) is obtained by fine-tuning a new uni-modal classifier with the corresponding trained uni-modal encoder. A larger spot size reflects a better multimodal performance. Our method is superior in both multimodal performance and all uni-modal performance.

Figure 1. Evaluation of scarcely informative modality cases and uni-modal encoder quality on CREMA-D and Kinetics Sounds.

5. Experimental Datasets

The authors evaluate their method on multiple datasets and modality combinations:

Item	Details
CREMA-D	Audio and vision dataset for emotion recognition.
Kinetics Sounds	Audio and vision dataset for action recognition.
UCF-101	RGB and optical flow dataset for human action recognition.
CMU-MOSI	Audio, vision, and text dataset for sentiment analysis.

6. Key Results

The proposed method consistently outperforms several existing imbalanced multimodal learning methods, including G-Blending, OGM-GE, Greedy, PMR, and AGM.

The results show that Diagnosing & Re-learning improves both multimodal performance and uni-modal representation quality.

Table 1: Comparison with imbalanced multimodal learning methods where bold and underline represent the best and second best respectively. Joint-training is the widely-used baseline with concatenation fusion and one multimodal loss function.

Method	CREMA-D (Audio/Vision)		Kinetics Sounds (Audio/Vision)		UCF-101 (RGB/Optical Flow)	
	Acc	Macro F1	Acc	Macro F1	Acc	Macro F1
Joint-training	67.47	67.80	65.04	65.12	80.41	79.40
G-Blending [20]	69.89	70.41	<u>68.60</u>	<u>68.64</u>	<u>81.73</u>	<u>80.84</u>
OGM-GE [15]	68.95	69.39	67.15	66.93	81.15	80.36
Greedy [25]	68.37	68.46	65.72	65.80	80.60	79.50
PMR [6]	68.55	68.99	65.62	65.36	81.36	80.37
AGM [9]	70.16	70.67	66.50	66.49	81.55	80.58
Ours	75.13	76.00	69.10	69.39	82.11	80.87

Table 2: Additional comparison with G-Blending [20] method. For a fair comparison, we also introduce the same uni-modal classifier and uni-modal cross-entropy loss function as G-Blending in the Ours[†] method.

Method	CREMA-D (Audio/Vision)		Kinetics Sounds (Audio/Vision)		UCF-101 (RGB/Optical Flow)	
	Acc	Macro F1	Acc	Macro F1	Acc	Macro F1
Joint-training	67.47	67.80	65.04	65.12	80.41	79.40
G-Blending [20]	69.89	70.41	68.60	68.64	81.73	80.84
Ours	75.13	76.00	69.10	69.39	82.11	80.87
Ours [†]	79.30	79.58	72.17	72.02	83.05	82.17

Tables 1 and 2. Comparison with imbalanced multimodal learning methods on CREMA-D, Kinetics Sounds, and UCF-101.

The paper also shows that the method remains effective in transformer-based frameworks and in settings with more than two modalities.

Table 3: (Left): Comparison with imbalanced multimodal learning methods on CMU-MOSI dataset with three modalities. The greedy method could not extend to cases with more than two modalities. * indicates that the original methods of OGM-GE and PMR only consider two modality cases, but we extend them while retaining the core uni-modal balancing strategy. **(Right): Comparison with imbalanced multimodal learning methods with Transformer-based backbone on CREMA-D dataset.**

Method	CMU-MOSI			CREMA-D		
	Transformer-based backbone (Audio/Vision/Text)			Transformer-based backbone (Audio/Vision)		
	Acc	Weighted F1	Macro F1	Acc	Weighted F1	Macro F1
Joint-training	76.96	76.76	75.68	68.55	68.88	69.33
G-Blending [20]	77.26	77.20	76.27	69.35	68.77	69.63
OGM-GE [15]	77.41*	77.17*	76.09*	70.70	70.67	71.18
Greedy [25]	/	/	/	68.41 (↓)	68.72 (↓)	69.17 (↓)
PMR [6]	77.55*	77.50*	76.58*	69.89	69.70	70.07
AGM [9]	77.26	77.07	76.02	69.22	69.34	69.61
Ours	77.99	78.09	77.37	71.64	71.66	72.12

Table 3. Comparison on CMU-MOSI with three modalities and on CREMA-D with a transformer-based backbone.

7. Scarcely Informative Modality Case

A key contribution of the paper is showing that existing methods can fail when one modality is noisy or scarcely informative. The authors modify the audio data in CREMA-D by adding white Gaussian noise. In this challenging setting, most compared methods experience a performance drop, while the proposed method still improves performance.

Table 4: Comparison with imbalanced multimodal learning methods on scarcely informative modality case. We modify the audio data of the CREMA-D dataset, adding extra white Gaussian noise to make it noisier and scarcely discriminative. All compared methods have a clear performance drop in this case. In contrast, our method still achieves considerable improvement in this challenging scenario.

Method	Scarcely informative modality case (Audio/Vision)		
	Acc	Weighted F1	Macro F1
Joint-training	65.86	66.03	66.39
G-Blending [20]	61.16 (↓)	60.45 (↓)	61.15 (↓)
OGM-GE [15]	62.63 (↓)	64.21 (↓)	65.07 (↓)
Greedy [25]	63.17 (↓)	62.99 (↓)	63.83 (↓)
PMR [6]	65.73 (↓)	64.81 (↓)	65.33 (↓)
AGM [9]	62.87 (↓)	62.28 (↓)	63.73 (↓)
Ours	69.49	69.72	70.28

Table 4. Results on the scarcely informative modality case with noisy audio data.

8. Representation Quality Analysis

The paper also visualizes uni-modal representation quality using t-SNE. Compared with joint training, the proposed method improves the separability of the weaker visual modality while preserving useful information from the audio modality.

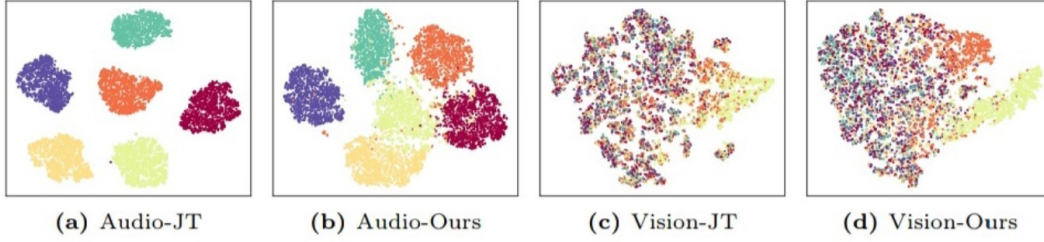


Fig. 3: Uni-modal representation visualization by t-SNE [12] on CREMA-D dataset. The categories are indicated in different colors. JT denotes for Joint-training.

Figure 3. t-SNE visualization of audio and vision representations on CREMA-D.

The purity gap and test accuracy curves further support the idea that the proposed method can diagnose learning states and improve balanced learning over time.

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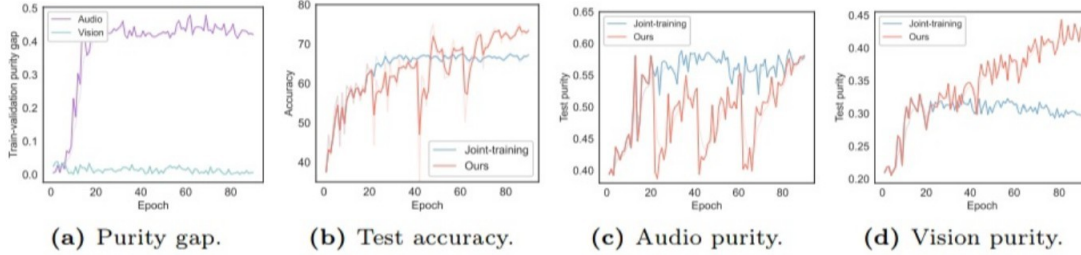


Fig. 4: (a): The purity gap between training and validation representation. (b): Changes in test accuracy during training. (c&d): The purity of test representation. All results are based on the CREMA-D dataset.

Figure 4. Purity gap, test accuracy, and test representation purity curves on CREMA-D.

9. Hyperparameter Sensitivity

The paper analyzes the sensitivity of two hyperparameters: the re-initialization strength controller and the diagnosing/re-learning frequency. The results show that the method generally improves performance across different settings.

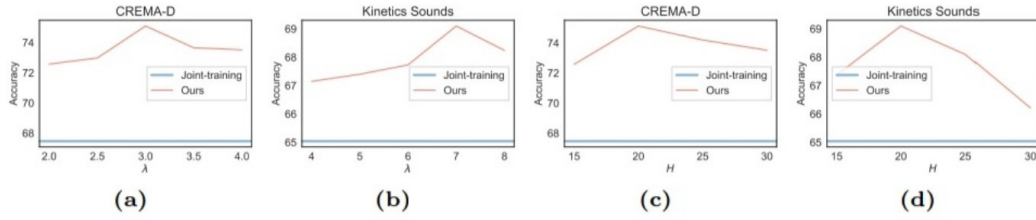


Fig. 5: Hyper-parameter sensitivity analysis of λ in Equation 5 and Diagnosing & Re-learning frequency H on CREMA-D and Kinetics Sounds datasets.

Figure 5. Hyperparameter sensitivity analysis on CREMA-D and Kinetics Sounds.

10. Research Gap

Research Gap

Although Diagnosing & Re-learning improves balanced multimodal learning, it is mainly evaluated on classification tasks. Its application to real-world multimodal CIR prototypes, especially systems combining noisy audio and facial expressions, remains a practical open gap.

11. Relevance to Our Multimodal CIR Project

This paper is directly useful for our project because our prototype focuses on audio and facial expressions. In real conditions, one modality may dominate the model decision. For example, audio may be noisy, or the face may be blurred or poorly illuminated.

The idea of diagnosing the learning state of each modality can help us design a more reliable prototype that does not blindly trust one modality over the other.

12. Key Takeaways

- Multimodal models may over-rely on one modality and under-learn others.
- Weak modality performance does not always mean the modality needs more training; it may be noisy or scarcely informative.
- Diagnosing & Re-learning estimates each modality learning state using representation separability.
- Soft re-initialization can reduce over-reliance, prevent over-training, and improve generalization.
- The method is relevant to audio-visual prototypes because audio and facial expressions may have different reliability levels in real-world scenarios.

13. Recommended Figures and Tables to Use

Item	Why It Matters	Suggested File Name
Figure 1	Shows scarcely informative modality and encoder quality comparison.	Fig1_Scarcely_Informative_Modality
Figure 2	Shows the full Diagnosing & Re-learning framework.	Fig2_Framework
Figure 3	Visualizes uni-modal representation quality using t-SNE.	Fig3_UniModal_tSNE
Figure 4	Shows purity gap, test accuracy, and representation purity.	Fig4_Purity_Gap_Analysis
Table 1	Main comparison with imbalance learning methods.	Table1_Method_Comparison
Table 4	Important evidence for noisy/scarcely informative modality case.	Table4_Noisy_Modality_Case